

LAPORAN PRAKTIKUM

PENGELOLAAN ADMINISTRASI & TEKNOLOGI INFORMASI

PROJECT-BASED: Mini Server Linux untuk Administrasi Sistem

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1. Studi Kasus

Sebuah institusi membutuhkan mini server Linux untuk kebutuhan internal. Server harus memenuhi kriteria: stabil saat boot, aman dari kegagalan disk, dapat dimonitor performanya, selalu up-to-date, dan siap digunakan oleh banyak pengguna. Kelompok kami bertugas sebagai tim administrator sistem untuk merancang, mengimplementasikan, dan mendemonstrasikan sistem tersebut.

Seluruh konfigurasi dilakukan menggunakan **konfigurasi manual** sebagai alat Infrastructure-as-Code untuk menjamin reproduisibilitas dan konsistensi deployment.

2. Arsitektur Sistem

Komponen	Detail
Host OS	Windows 10/11
Control Node	WSL (Ubuntu) - menjalankan konfigurasi manual
Hypervisor	Oracle VirtualBox
Guest OS	Ubuntu Server 24.04.4 LTS (Headless)
Kernel	6.8.0-110-generic (x86_64)
Boot Loader	GRUB 2.12
Jaringan - NAT	Akses internet & update paket
Jaringan - Host-Only	192.168.56.101 (manajemen server & Samba)
Disk Primer	sda - 20 GB (sistem /, /boot)
Disk Sekunder	sdb - 20 GB -> sdb1 /data (ext4)
User	adminuser (SSH + Samba)
Timezone	Asia/Jakarta

Gambar 1 - Login & System Info Ubuntu Server

```
mini-server login: adminuser
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Apr 24 06:59:41 PM UTC 2026

System load:          0.08
Usage of /:           47.7% of 9.75GB
Memory usage:        13%
Swap usage:          0%
Processes:           129
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fec9:63e7

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

adminuser@mini-server:~$ systemctl status multipathd
* multipathd.service - Device-Mapper Multipath Device Controller
   Loaded: loaded (/usr/lib/systemd/system/multipathd.service; disabled; preset: enabled)
   Active: inactive (dead) since Fri 2026-04-24 18:56:52 UTC; 3min 7s ago
   Duration: 2min 49.792s
   TriggeredBy: ● multipathd.socket
   Process: 9465 ExecStart=/sbin/multipathd -d -s (code=exited, status=0/SUCCESS)
   Main PID: 9465 (code=exited, status=0/SUCCESS)
   Status: "shutdown"
   CPU: 406ms

Apr 24 18:54:01 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:54:01 mini-server systemd[1]: multipathd.service: Consumed 1.155s CPU time, 22.1M memory peak, 0B memory swap peak.
Apr 24 18:54:01 mini-server systemd[1]: Starting multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:54:02 mini-server multipathd[9465]: multipathd v0.9.4: start up
Apr 24 18:54:02 mini-server multipathd[9465]: reconfigure: setting up paths and maps
Apr 24 18:54:02 mini-server systemd[1]: Started multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:56:52 mini-server multipathd[9465]: multipathd: shut down
Apr 24 18:56:52 mini-server systemd[1]: Stopping multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:56:52 mini-server systemd[1]: multipathd.service: Deactivated successfully.
Apr 24 18:56:52 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
adminuser@mini-server:~$
```

3. Implementasi Teknis

3.1 Boot & Service Management

Boot Loader: GRUB 2.12 - teridentifikasi melalui `grub-install --version`.

Kernel: Linux 6.8.0-110-generic - teridentifikasi melalui `uname -r`.

Service yang dinonaktifkan: `multipathd.service` (Device-Mapper Multipath Device Controller). Service ini tidak diperlukan karena VM tidak menggunakan multipath storage.

Custom Startup Service: `hello-boot.service` - service sederhana yang mencatat pesan ke journal saat sistem boot, membuktikan kemampuan membuat dan mengaktifkan unit systemd.

Gambar 2 - Status multipathd (disabled)

```
mini-server login: adminuser
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
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Usage of /:           47.7% of 9.75GB
Memory usage:         13%
Swap usage:           0%
Processes:            129
Users logged in:      1
IPV4 address for enp0s3: 10.0.2.15
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adminuser@mini-server:~$ systemctl status multipathd
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     Duration: 2min 49.752s
   TriggeredBy: ● multipathd.socket
   Process: 9465 ExecStart=/sbin/multipathd -d -s (code=exited, status=0/SUCCESS)
   Main PID: 9465 (code=exited, status=0/SUCCESS)
   Status: "shutdown"
     CPU: 406ms

Apr 24 18:54:01 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:54:01 mini-server systemd[1]: multipathd.service: Consumed 1.155s CPU time, 22.1M memory peak, 0B memory swap peak.
Apr 24 18:54:01 mini-server systemd[1]: Starting multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:54:02 mini-server multipathd[9465]: multipathd v0.9.4: start up
Apr 24 18:54:02 mini-server multipathd[9465]: reconfigure: setting up paths and maps
Apr 24 18:54:02 mini-server systemd[1]: Started multipathd.service - Device-Mapper Multipath Device Controller.
Apr 24 18:56:52 mini-server multipathd[9465]: multipathd: shut down
Apr 24 18:56:52 mini-server systemd[1]: Stopping multipathd.service - Device-Mapper Multipath Device Controller...
Apr 24 18:56:52 mini-server systemd[1]: multipathd.service: Deactivated successfully.
Apr 24 18:56:52 mini-server systemd[1]: Stopped multipathd.service - Device-Mapper Multipath Device Controller.
adminuser@mini-server:~$
```

Gambar 3 - Status hello-boot.service (enabled, executed)

```
adminuser@mini-server:~$ systemctl status hello-boot.service
* hello-boot.service - Custom Welcome Boot Service
   Loaded: loaded (/etc/systemd/system/hello-boot.service; enabled; preset: enabled)
   Active: inactive (dead) since Fri 2026-04-24 18:54:54 UTC; 6min ago
   Main PID: 11770 (code=exited, status=0/SUCCESS)
     CPU: 8ms

Apr 24 18:54:54 mini-server systemd[1]: Starting hello-boot.service - Custom Welcome Boot Service...
Apr 24 18:54:54 mini-server systemd[1]: hello-boot.service: Deactivated successfully.
Apr 24 18:54:54 mini-server systemd[1]: Finished hello-boot.service - Custom Welcome Boot Service.
adminuser@mini-server:~$ _
```

3.2 Manajemen Disk & Partisi

Sistem menggunakan **2 disk virtual**:

- **sda** (20 GB) - Disk sistem: berisi /boot (sda2, 1.8 GB), LVM root / (sda3 -> ubuntu--vg-ubuntu--lv, 10 GB)
- **sdb** (20 GB) - Disk data: partisi sdb1 di-mount ke /data dengan filesystem ext4

Partisi data dipisahkan agar kerusakan disk sistem tidak mempengaruhi data pengguna, dan sebaliknya.

Gambar 4 - Output lsblk (struktur partisi)

```
adminuser@mini-server:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0      0   20G  0 disk
├─sda1                              8:1      0    1M  0 part
├─sda2                              8:2      0   1.8G  0 part /boot
└─sda3                              8:3      0  18.2G  0 part
   └─ubuntu--vg-ubuntu--lv 252:0      0    10G  0 lvm  /
sdb                                  8:16     0   20G  0 disk
└─sdb1                              8:17     0   20G  0 part /data
sr0                                  11:0     1 1024M  0 rom
adminuser@mini-server:~$
```

3.3 Keamanan Data (Backup & Recovery)

Kelompok memilih opsi **mekanisme cadangan data manual** menggunakan *rsync* dan *cron*.

Konfigurasi Cron:

```
0 2 * * * rsync -a /data/ /backup/ > /var/log/data_backup.log 2>&1
```

Backup otomatis berjalan setiap hari pukul 02:00 WIB.

Simulasi Kegagalan & Pemulihan:

1. File *Hi.txt* dibuat di */data/share/* dari PC Windows melalui Samba
2. File dihapus: `sudo rm /data/share/Hi.txt`
3. Restore dari backup: `sudo rsync -a /backup/share/ /data/share/`
4. Verifikasi: `cat /data/share/Hi.txt` - data berhasil dipulihkan

Gambar 5 - Crontab entry untuk backup

```
adminuser@mini-server:~$ sudo crontab -l -u root
#Ansible: Backup Data Partition
0 2 * * * rsync -a /data/ /backup/ > /var/log/data_backup.log 2>&1
adminuser@mini-server:~$
```

Gambar 6 - Simulasi delete & recovery

```
adminuser@mini-server:/data/share$ sudo rsync -a /data/ /backup/
adminuser@mini-server:/data/share$ ls
Hi.txt
adminuser@mini-server:/data/share$ cat Hi.txt
Hello, this is the host PC!
adminuser@mini-server:/data/share$ sudo rm /data/share/Hi.txt
adminuser@mini-server:/data/share$ sudo rsync -a /backup/share/ /data/share/
adminuser@mini-server:/data/share$ cat /data/share/Hi.txt
Hello, this is the host PC!
adminuser@mini-server:/data/share$ _
```

3.4 Monitoring Sistem

Tools yang diinstall: **htop**, **sysstat** (sar/iostat), dan **stress**.

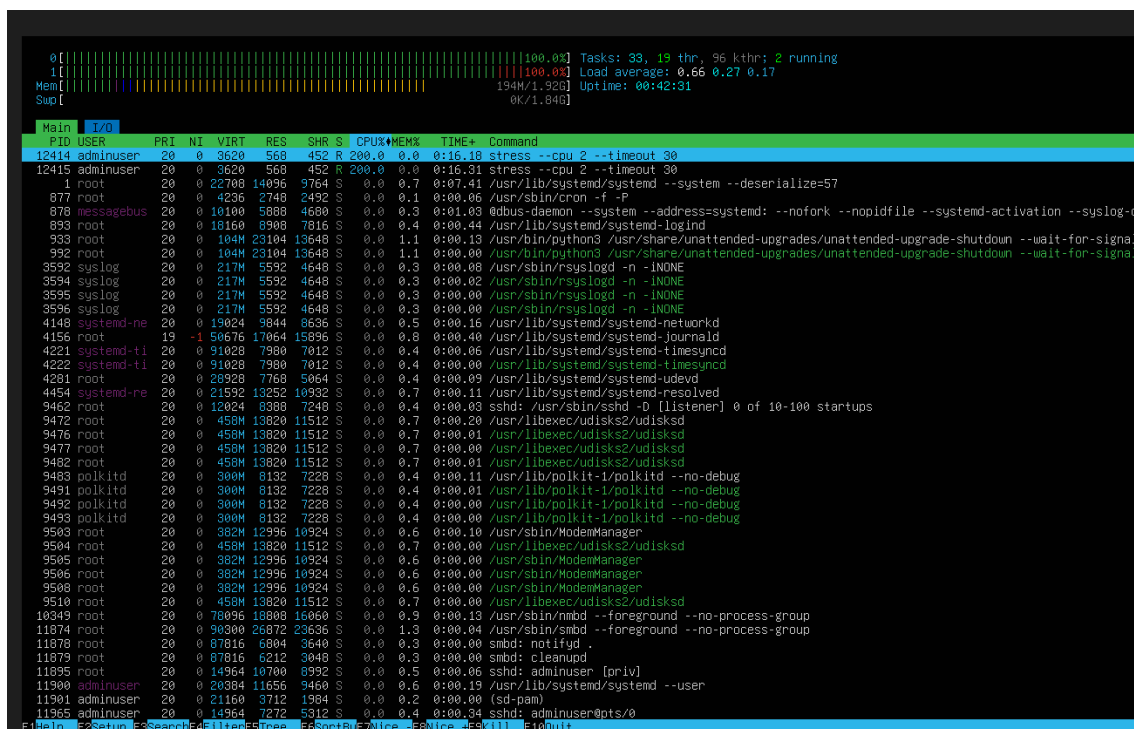
Simulasi beban tinggi:

```
stress --cpu 2 --timeout 30
```

Perintah ini membebani 2 CPU worker selama 30 detik.

Analisis: Saat stress test berjalan, kedua core CPU mencapai 100% utilisasi. Proses *stress* mengkonsumsi masing-masing 200% CPU (per-core). Load average naik dari ~0.08 ke ~0.66. Penggunaan RAM tetap stabil di ~194 MB / 1.9 GB karena stress test hanya membebani CPU. Setelah timeout berakhir, CPU kembali normal dalam beberapa detik.

Gambar 7 - htop saat stress test (CPU 100%)



Gambar 8 - htop setelah stress berakhir



3.5 Update Sistem & Keamanan Dasar

Update: Sistem di-update melalui *apt update* && *apt upgrade -y*.

Firewall (UFW):

Port/Service	Action	Keterangan
22/tcp (SSH)	ALLOW	Akses remote administrasi
Samba	ALLOW	File sharing ke Windows
23/tcp (Telnet)	DENY	Ditutup karena tidak aman (plaintext)

Gambar 9 - apt update

```
adminuser@mini-server:~$ sudo apt update
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
adminuser@mini-server:~$
```

Gambar 10 - ufw status

```
adminuser@mini-server:~$ sudo ufw status
[sudo] password for adminuser:
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Samba ALLOW Anywhere
23/tcp DENY Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
Samba (v6) ALLOW Anywhere (v6)
23/tcp (v6) DENY Anywhere (v6)

adminuser@mini-server:~$ _
```

3.6 Layanan Tambahan - Samba File Sharing (Ops B)

Kelompok memilih **Ops B: File Sharing** menggunakan Samba.

Konfigurasi share: Folder `/data/share` di-share sebagai *PraktikumShare* dengan autentikasi user-level (adminuser).

Akses dari Windows: Melalui File Explorer -> `\\192.168.56.101`

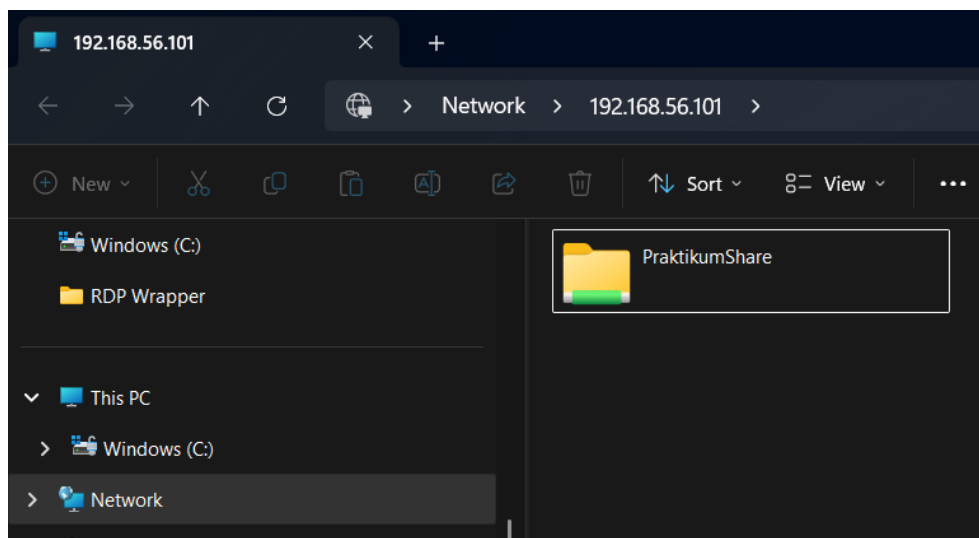
Pengujian: File *Hi.txt* berhasil dibuat dari Windows, terbaca di server Linux, dan sebaliknya - membuktikan file sharing dua arah berfungsi.

Gambar 11 - Status Samba (smbd active)

```
adminuser@mini-server:~$ systemctl status smbd
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-04-24 18:54:56 UTC; 10min ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
   Main PID: 11874 (smbd)
    Status: "smbd: ready to serve connections..."
     Tasks: 3 (limit: 2263)
    Memory: 7.5M (peak: 7.9M)
       CPU: 71ms
    CGroup: /system.slice/smbd.service
            └─11874 /usr/sbin/smbd --foreground --no-process-group
              └─11878 "smbd: notifyd"
                └─11879 "smbd: cleanupd"

Apr 24 18:54:56 mini-server systemd[1]: Starting smbd.service - Samba SMB Daemon...
Apr 24 18:54:56 mini-server (smbd)[11874]: smbd.service: Referenced but unset environment variable evaluates to an empty string: SMBOPTIONS
Apr 24 18:54:56 mini-server systemd[1]: Started smbd.service - Samba SMB Daemon.
adminuser@mini-server:~$
```

Gambar 12 - Akses dari Windows Explorer



Gambar 13 - Isi folder PraktikumShare

A screenshot of a Windows File Explorer window showing the contents of the 'PraktikumShare' folder. The address bar shows the path 'Network > 192.168.56.101 > PraktikumShare'. The table below represents the data shown in the file list.

Name	Date modified	Type	Size
Hi	4/25/2026 2:09 AM	Text Document	1 KB

Gambar 14 - Verifikasi isi file dari server

```
adminuser@mini-server:/data/share$ cat Hi.txt
Hello, this is the host PC!adminuser@mini-server:/data/share$ ls
Hi.txt
adminuser@mini-server:/data/share$ _
```

4. Masalah & Solusi

No	Masalah	Solusi
1	Timezone VM default UTC, cron tidak sinkron dengan waktu presentasi	Mengubah timezone ke Asia/Jakarta: <code>timedatectl set-timezone Asia/Jakarta</code>
2	Password Samba & Linux terpisah - akses ditolak dari Windows	Menjalankan <code>smbpasswd -a adminuser</code> untuk sinkronisasi kredensial
3	multipathd aktif tanpa multipath storage (warning di log)	Menonaktifkan service: <code>systemctl disable --now multipathd</code>
4	Backup directory /backup belum ada saat pertama kali rsync	Directory /backup harus dibuat secara manual sebelum konfigurasi cron

5. Analisis Administrator

Dari implementasi ini, kami mengambil beberapa pelajaran penting sebagai administrator sistem:

- **Automasi adalah kunci:** Menggunakan konfigurasi manual memungkinkan seluruh konfigurasi server direproduksi dalam hitungan menit, mengurangi human error.
- **Pemisahan disk** antara sistem dan data adalah praktik terbaik untuk isolasi kegagalan.
- **Backup terjadwal** dengan rsync + cron memberikan jaring pengaman minimum tanpa memerlukan solusi RAID yang lebih kompleks.
- **Firewall** harus dikonfigurasi secara proaktif - menutup port yang tidak perlu (seperti Telnet/23) mencegah attack surface yang tidak diperlukan.
- **Monitoring** dengan htop memberikan visibilitas real-time yang sangat berguna untuk diagnosis masalah performa.
- **Samba** membuktikan Linux dapat berinteroperasi dengan ekosistem Windows secara seamless untuk kebutuhan file sharing.

6. Kesimpulan

Kelompok kami telah berhasil merancang, mengimplementasikan, dan mendemonstrasikan sebuah mini server Linux menggunakan Ubuntu Server 24.04 LTS di lingkungan VirtualBox. Seluruh spesifikasi tugas praktikum telah dipenuhi:

1. Boot loader (GRUB 2.12) dan kernel (6.8.0-110-generic) teridentifikasi, service dikelola melalui systemd.
2. Dua disk dikonfigurasi dengan partisi sistem dan data terpisah.
3. Mekanisme backup rsync + cron diimplementasikan dan recovery berhasil dibuktikan.
4. Monitoring CPU dilakukan dengan htop dan stress test.
5. Sistem di-update dan firewall UFW dikonfigurasi.
6. Samba file sharing berhasil diakses dari Windows.

Seluruh konfigurasi otomatisasi menggunakan konfigurasi manual, menjadikan sistem ini sepenuhnya reproducible dan siap untuk deployment ulang kapan saja.